



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam: Trimester: Summer 2025

Course Code: EEE 2123; Course Title: Electronics
Total Marks: 40; Duration: 2hr

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

There are six (6) questions in this paper. Answer all of them.

1. The MOSFET in the following figure has $V_t = 1\text{ V}$, $k'_n \frac{W}{L} = 1\text{ mA/V}^2$. What is the largest value that R_D can have while the transistor remains in saturation mode? [7]

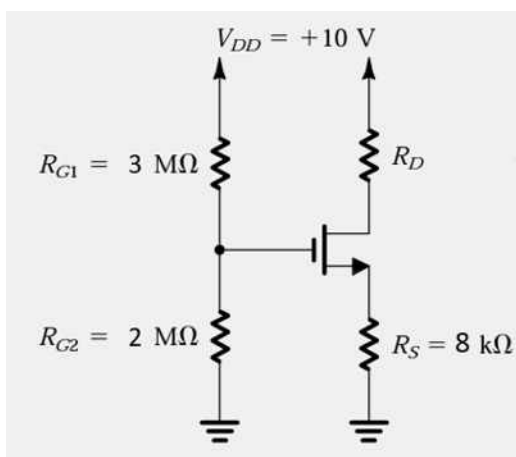


fig: Q1

2. Determine the values of R_D and R_S so that the transistor operates at $I_D = 0.3\text{ mA}$ and $V_D = +0.4\text{ V}$. The NMOS transistor has $V_t = 1\text{ V}$, $\mu_n C_{ox} = 60\text{ }\mu\text{A/V}^2$, $W/L = 40$ [7]

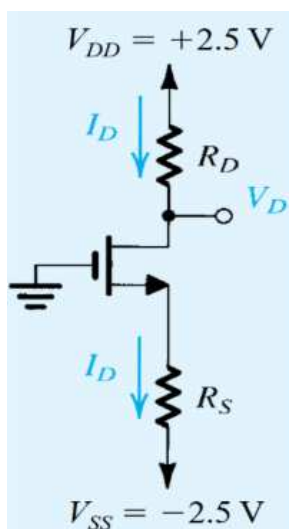


fig: Q2

3. For the circuit shown in Figure 4, the transistor parameters are $R_{B1} = R_{B2} = 50\text{k}\Omega$, $\beta = 100$. Find I_B , I_C , I_E , and V_{CE} . [7]

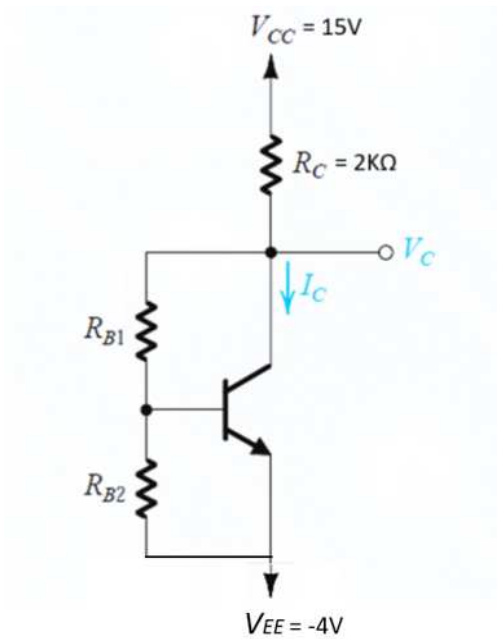


fig: Q3

4. Find all node voltages and branch currents in the following circuit, $\beta = 130$ [7]

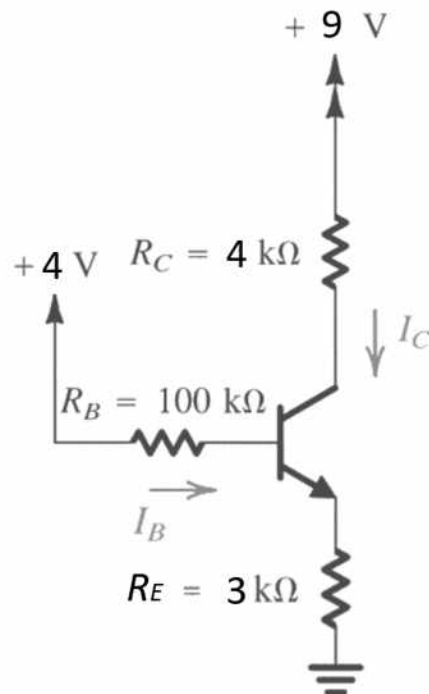


fig: Q4

5. (a) The **NMOS** network of a random digital system is given below:

[3]

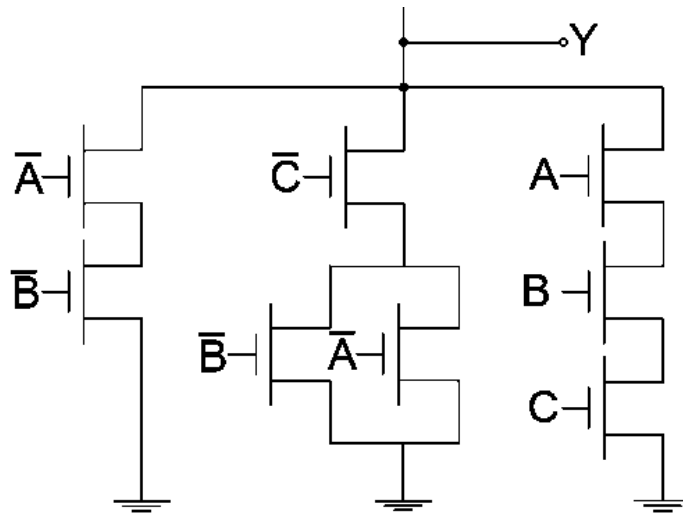


fig: Q5(a)

Draw the **PMOS** network and derive the corresponding output expression.

- (b) Implement the following logic function using CMOS:

[3]

$$f = (a + bc) \cdot (e + d) + gh$$

6. (a) Implement the following equation using Op-Amps. Clearly mention the values of all resistances and capacitances used in the design.

[4]

$$v_o = 5v_{in} + 10 \int (-6 \int v_{in} dt + 7 \frac{dv_{in}}{dt}) dt$$

- (b) You are given (**$V_i = \sin t$**) as the only input. You have to get the following output expression using op-amps, **Note: You cannot use differentiator.**

[2]

$$v_o = 3 \sin t - 5 \cos t$$